TECHNICAL PROJECT REPORT

House Price Prediction

Using Regression Models with Streamlit Deployment

An end-to-end machine learning pipeline for predicting residential property prices in Melbourne, comparing four regression algorithms and deploying the best performer as an interactive Streamlit web application.

Dataset: Melbourne Housing Snapshot (Kaggle)

Models: Linear Regression · Ridge Regression · Random Forest · Gradient Boosting

Deployment: Streamlit Web Application

Executive Summary

This project presents an end-to-end machine learning solution for predicting residential property prices in Melbourne, Australia. The work compares four regression algorithms—Linear Regression, Ridge Regression, Random Forest Regressor, and Gradient Boosting Regressor—on the public Melbourne Housing Snapshot dataset from Kaggle.

After systematic data preprocessing, outlier removal, feature scaling, and one-hot encoding of categorical variables, the Gradient Boosting Regressor was identified as the best-performing model based on R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The final model was serialized with joblib and deployed as an interactive Streamlit web application, enabling real-time price predictions from user-provided property attributes.

The report documents the entire workflow—from exploratory data analysis and feature engineering through model selection, evaluation, and deployment—and concludes with recommendations for future extensions, such as geospatial enrichment and API-based deployment.

1. Introduction

The real estate market is one of the most dynamic sectors in any economy, where accurate property valuation plays a crucial role for buyers, sellers, investors, and lenders. Traditional valuation methods often rely on manual appraisal and subjective judgment, which can introduce inconsistency, bias, and significant delays.

With the rise of data-driven decision-making and machine learning, predictive modeling offers a more reliable, scalable, and reproducible approach to property valuation. Supervised regression models can capture complex, non-linear relationships between property characteristics and market prices, producing objective estimates that complement expert judgment.

1.1 Project Objectives

- Build a supervised regression model that predicts house prices from structured property features.
- Compare four regression algorithms and select the best performer based on standardized metrics.
- Deploy the trained model as an interactive, browser-based application using Streamlit.
- Provide comprehensible information about the features that most influence predicted price.

1.2 Scope

The scope of this project is limited to tabular regression on the Melbourne Housing Snapshot dataset. Time-series dynamics, external economic indicators, and image-based features (e.g., property photos) are out of scope. The deployed application is a single-user prototype intended to demonstrate the end-to-end workflow.

2. Dataset

The dataset used in this project is the Melbourne Housing Snapshot, publicly available on Kaggle. It contains detailed information about residential properties in Melbourne, including physical characteristics, geographic location, and transaction details.

Source: Melbourne Housing Snapshot on Kaggle

2.1 Key Features

Feature	Description
Rooms	Number of bedrooms in the property
Distance	Distance from the central business district (km)
Bathroom	Number of bathrooms
Car	Number of car parking spaces
Landsize	Size of the property land (m^2)
Building Area	Total built-up area of the property (m^2)
YearBuilt	Year the property was constructed
Suburb / Region	Categorical geographic identifiers
Type, Method, SellerG	Categorical transactional attributes

2.2 Target Variable

Price — the selling price of the property in Australian dollars. This is a continuous numerical variable, making the task a supervised regression problem.

[29]:

	Suburb	Address	Rooms	Type	Price	Method	SellerG	Date	Distance	Postcode	...	Bathroom	Car	Landsize	BuildingArea	YearBuilt	CouncilArea
0	Abbotsford	85 Turner St	2	h	1480000.0	S	Biggin	3/12/2016	2.5	3067.0	...	1.0	1.0	202.0	NaN	NaN	Yarra
1	Abbotsford	25 Bloomburg St	2	h	1035000.0	S	Biggin	4/02/2016	2.5	3067.0	...	1.0	0.0	156.0	79.0	1900.0	Yarra
2	Abbotsford	5 Charles St	3	h	1465000.0	SP	Biggin	4/03/2017	2.5	3067.0	...	2.0	0.0	134.0	150.0	1900.0	Yarra
3	Abbotsford	40 Federation La	3	h	850000.0	PI	Biggin	4/03/2017	2.5	3067.0	...	2.0	1.0	94.0	NaN	NaN	Yarra
4	Abbotsford	55a Park St	4	h	1600000.0	VB	Nelson	4/06/2016	2.5	3067.0	...	1.0	2.0	120.0	142.0	2014.0	Yarra

5 rows × 21 columns

3. Methodology

The project follows a standard, reproducible machine learning workflow consisting of six sequential stages:

1. **Data preprocessing** — handling missing values, removing outliers, and selecting features.
2. **Feature scaling and transformation**—standardizing numerical features and encoding categoricals.

3. **Data splitting** — producing training and testing subsets for unbiased evaluation.
4. **Model development** — training four regression algorithms with hyperparameter tuning.
5. **Model evaluation** — comparing models using R^2, MAE, and RMSE.
6. **Deployment** — packaging the best model into an interactive Streamlit application.

4. Data Preprocessing

Raw data rarely arrives in a form suitable for modeling. The preprocessing pipeline applied in this project addressed three primary issues: missing values, outliers, and mixed data types.

4.1 Handling Missing Values

Numerical features containing missing values were imputed using the median, which is robust to the positive skew typical of real-estate data. Columns deemed irrelevant or redundant (e.g., high-cardinality identifiers, fields with excessive missingness) were dropped from the modeling set.

```
Suburb              0
Address             0
Rooms              86
Type                0
Price             232
Method              0
SellerG             0
Date                0
Distance          246
Postcode          162
Bedroom2           99
Bathroom          143
Car               200
Landsize           24
BuildingArea     6454
YearBuilt        5381
CouncilArea      1369
Lattitude         139
Longtitude        174
Regionname          0
Propertycount     359
dtype: int64
```

```
145]: Suburb         0
      Rooms          0
      Type           0
      Price          0
      Method         0
      SellerG        0
      Distance       0
      Bathroom       0
      Car            0
      Landsize       0
      BuildingArea   0
      CouncilArea    0
      Regionname     0
      dtype: int64
```

```
No null values right now
```

Before and after removing missing values

4.2 Outlier Removal (Z-Score Method)

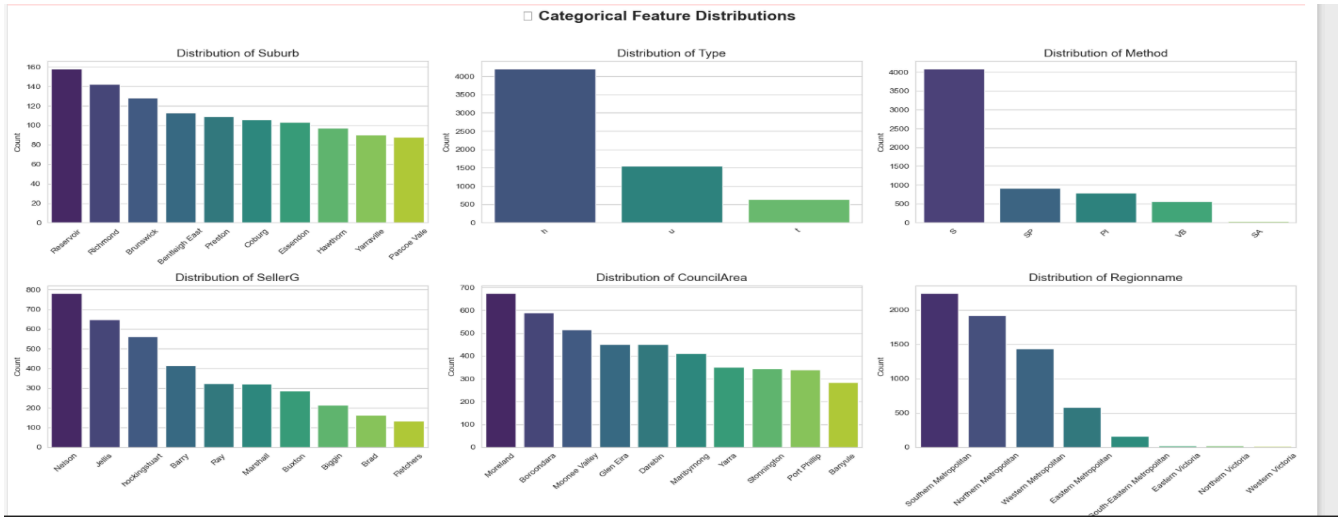
Outliers in continuous features such as price, land size, and building area were detected using the Z-score method with a threshold of |Z| > 3. Records flagged as extreme were removed to prevent them from distorting model training, particularly for loss functions sensitive to large residuals.

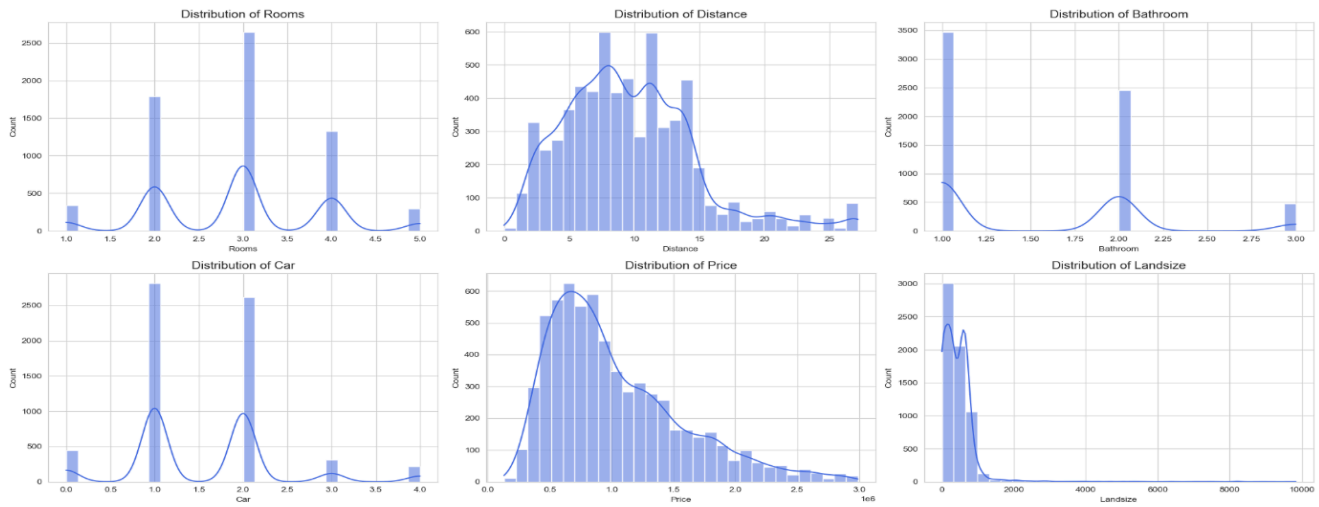
4.3 Encoding Categorical Variables

Nominal categorical variables such as Suburb, Type, and SellerG were transformed using one-hot encoding to produce numeric inputs compatible with all four regression algorithms. Ordinal encoding was avoided to prevent the imposition of a spurious ordering on unordered categories.

5. Exploratory Data Analysis

5.1 Distribution of Numerical Features



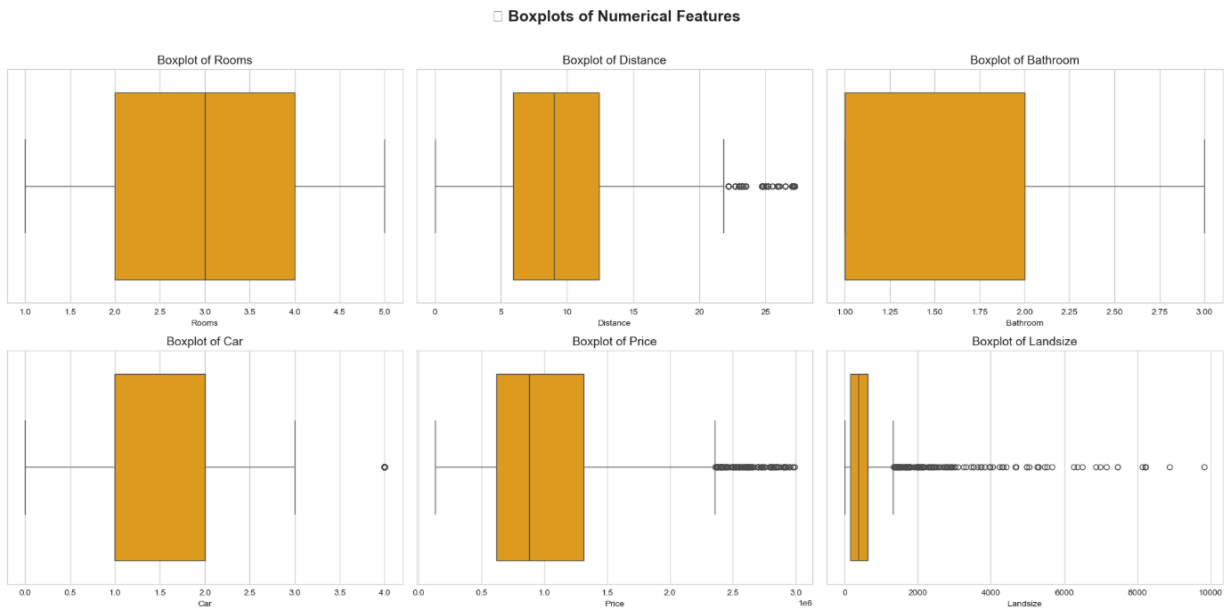


Discrete / count features (multimodal): Rooms, bathrooms, and cars exhibit distributions typical of count data, with values restricted to integer or half-integer steps.

Continuous features (skewed): Distance, Price, and Landsize display positive (right) skewness, with Landsize showing the most extreme skew.

5.2 Box-Plot Analysis

Central tendency: Median values for most features cluster at lower counts—3 rooms, 1 bathroom, and 1 car are typical.

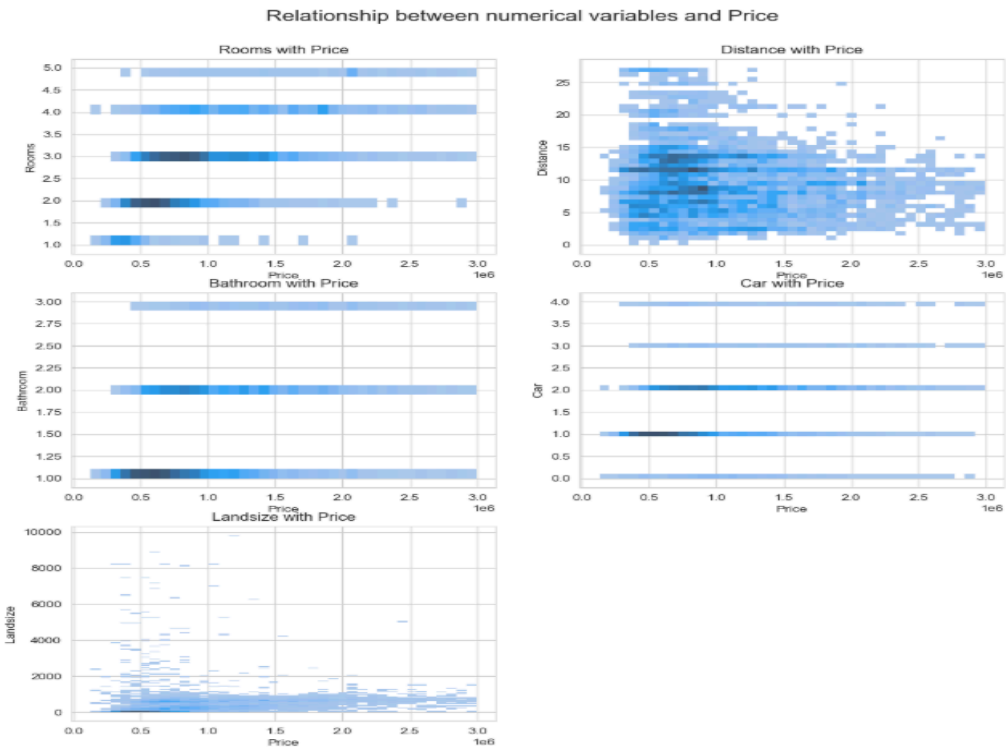


High-end outliers strongly affect price and especially land size, reinforcing the positive skew observed in the distribution plots. Distance also shows a notable number of high-end outliers.

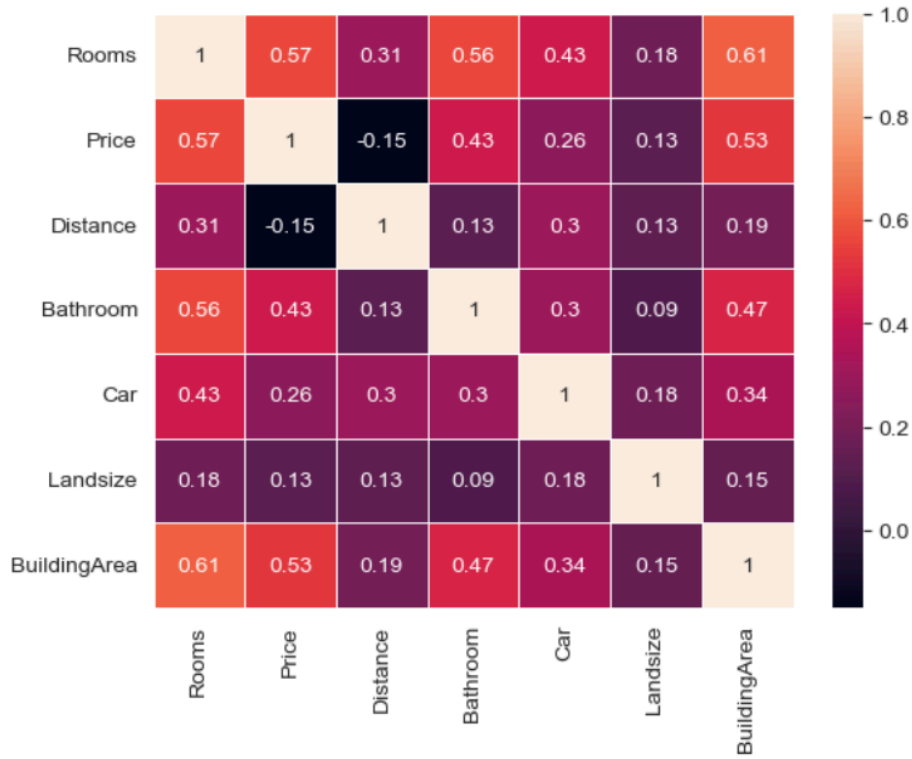
Feature spread: Features such as Rooms and Bathroom have narrow interquartile ranges relative to their maxima, while Price and Land size distributions are dominated by their outlier spread.

5.3 Correlation Structure

Correlation analysis reveals that rooms, prices, bathrooms, cars, and building areas form a cluster of interrelated attributes that collectively describe the size and utility of a property. Distance and land size behave relatively independently of this cluster, suggesting they contribute complementary signals to the predictive task.


Relationship between numerical variables and Price
Rooms with Price
Distance with Price
Bathroom with Price
Car with Price
Landsize with Price
Rooms
Distance
Bathroom
Car
Landsize
Price
1e6

[77]: <Axes: >


Rooms 1 0.57 0.31 0.56 0.43 0.18 0.61
Price 0.57 1 -0.15 0.43 0.26 0.13 0.53
Distance 0.31 -0.15 1 0.13 0.3 0.13 0.19
Bathroom 0.56 0.43 0.13 1 0.3 0.09 0.47
Car 0.43 0.26 0.3 0.3 1 0.18 0.34
Landsize 0.18 0.13 0.13 0.09 0.18 1 0.15
BuildingArea 0.61 0.53 0.19 0.47 0.34 0.15 1
Rooms
Price
Distance
Bathroom
Car
Landsize
BuildingArea
1.0
0.8
0.6
0.4
0.2
0.0

6. Feature Scaling and Transformation

Because features such as land size, distance, and building area span very different numerical ranges, StandardScaler was applied to standardize them to zero mean and unit variance. Standardization ensures that all features contribute on a comparable scale, preventing linear and regularized models from being dominated by high-magnitude features.

7. Train-Test Split

The preprocessed dataset was divided into training and testing subsets using a stratified, reproducible random split:

- **Training set:** 80% of observations—used to fit model parameters.
- **Testing set:** 20% of observations—held out for an unbiased performance evaluation.

A fixed random seed was used throughout to ensure that results are reproducible across runs and environments.

8. Model Development

Four regression algorithms were trained and compared to identify the best performer for this problem:

Algorithm	Rationale
Linear Regression	Simple, interpretable baseline; useful for quantifying linear signal.
Ridge Regression	The L2-regularized linear model controls for overfitting on one-hot encoded suburb features.
Random Forest Regressor	Tree ensemble that captures non-linear interactions with minimal tuning.
Gradient Boosting Regressor	Sequential boosting that typically achieves state-of-the-art tabular performance.

```
📊 Random Forest Results:
   - Mean Squared Error (MSE): 56522398201.9588
   - Root Mean Squared Error (RMSE): 237744.3968
   - R² Score: 0.8126
--------------------------------------------------
📊 Linear Regression Results:
   - Mean Squared Error (MSE): 71768170020.6966
   - Root Mean Squared Error (RMSE): 267895.8193
   - R² Score: 0.7620
--------------------------------------------------
📊 Ridge Regression Results:
   - Mean Squared Error (MSE): 69509023337.2332
   - Root Mean Squared Error (RMSE): 263645.6397
   - R² Score: 0.7695
--------------------------------------------------
📊 Gradient Boosting Results:
   - Mean Squared Error (MSE): 59729713535.3223
   - Root Mean Squared Error (RMSE): 244396.6316
   - R² Score: 0.8019
--------------------------------------------------
```

8.1 Training Workflow

7. Defined the feature matrix X and target vector y (Price).
8. Performed hyperparameter tuning using cross-validated grid search on the training set.
9. Fitted each candidate model on the training set with its best hyperparameters.
10. Serialized the final selected model with joblib for downstream deployment.

9. Model Evaluation

Model performance was evaluated on the held-out test set using three complementary regression metrics:

- **R^2 Score:** the proportion of variance in Price explained by the model; higher is better.
- **Mean Absolute Error (MAE):** the average absolute difference between predicted and actual prices; lower is better.
- **Root Mean Squared Error (RMSE):** penalizes large errors more heavily than MAE; lower is better.

```
📊 Random Forest Results:
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   - R² Score: 0.8019
--------------------------------------------------
```
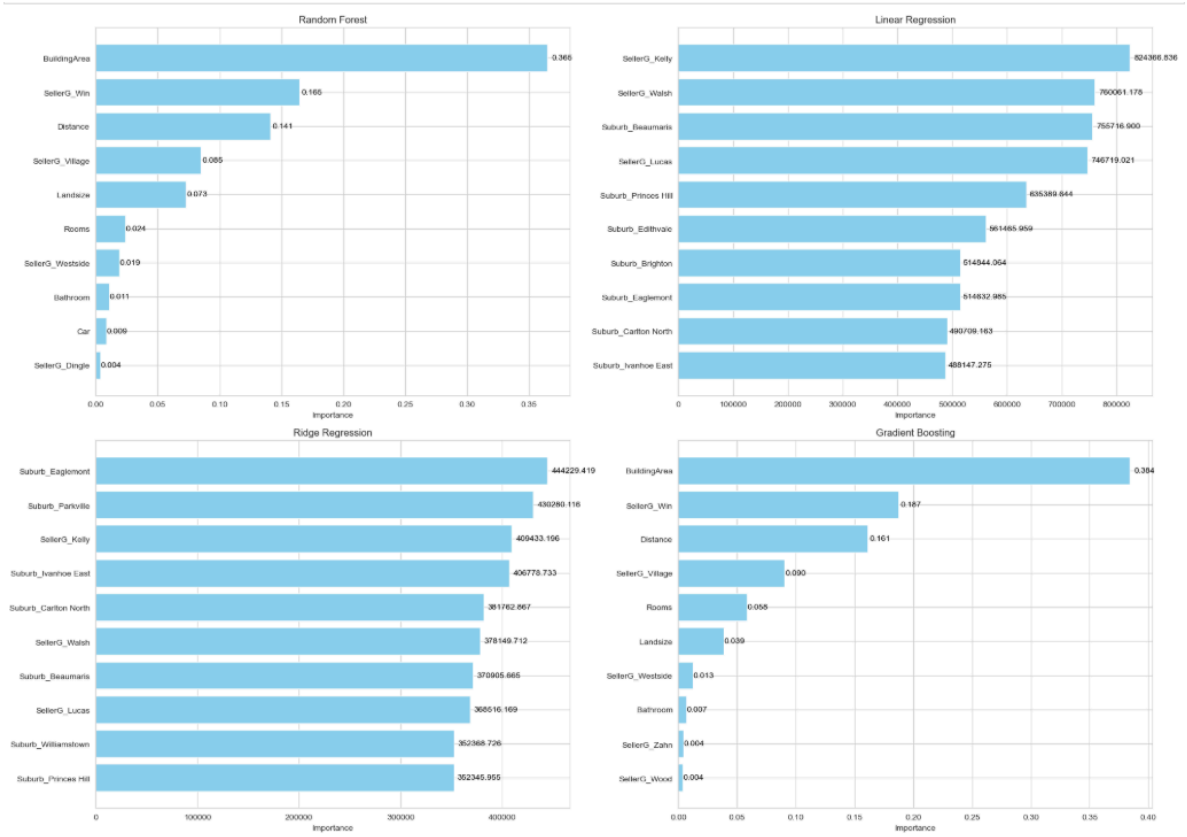

The Gradient Boosting Regressor achieved the highest R^2 and the lowest RMSE across the four candidates, making it the selected model for deployment. Its ability to capture non-linear feature interactions without explicit feature engineering proved particularly valuable on this dataset.

9.1 Feature Importance Across Models

Ensemble models (Random Forest, Gradient Boosting): heavily prioritize BuildingArea and Distance. Categorical features such as Seller and Agency receive moderate-to-high importance, while Rooms, Bathroom, and Landsize contribute comparatively less.

Linear models (linear regression, ridge regression): overwhelmingly prioritize location (suburb) through their large one-hot-encoded coefficients. This reflects the linear model's reliance on explicit location indicators to capture price variation.



10. Prediction Results

The trained gradient boosting model was used to generate predictions on unseen property data. The results align with observed market intuition, and the feature importance analysis highlights the following dominant predictors:

- **Building Area:** largest positive influence on predicted price.
- **Rooms:** strong positive correlation with price.
- **Distance:** negative correlation — properties closer to the central business district command higher prices.
- **Bathroom and Car:** moderate positive influence.

A feature importance plot generated from the final model provides an accessible visual summary of these contributions, supporting both model interpretability and stakeholder communication.

11. Deployment with Streamlit

To make the predictive model accessible to non-technical end users, it was deployed as an interactive web application using Streamlit. Streamlit was selected for its minimal boilerplate, native support for Python data-science libraries, and rapid iteration cycle.

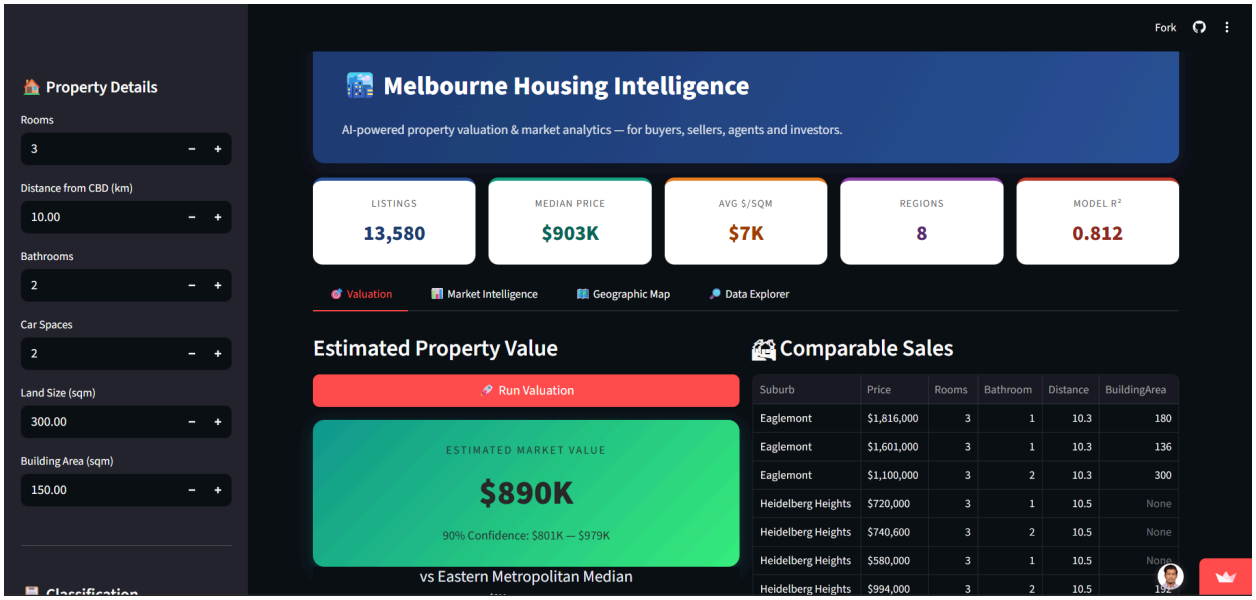
11.1 Deployment Steps

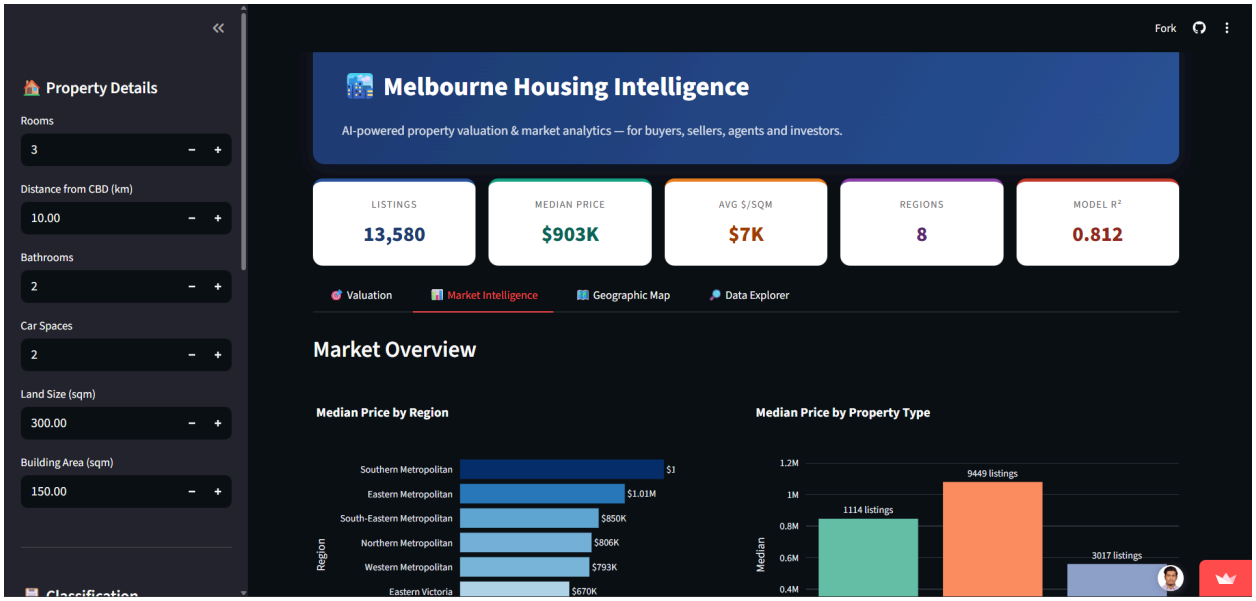
- **Model loading:** the serialized .pkl model is loaded at application start-up using joblib.
- **User input interface:** a sidebar form collects property attributes (rooms, bathrooms, land size, etc.) from the user.
- **Real-time prediction:** upon form submission, the application returns the estimated price instantly.
- **Visualization:** Feature-importance charts and summary performance metrics are rendered directly in the browser.

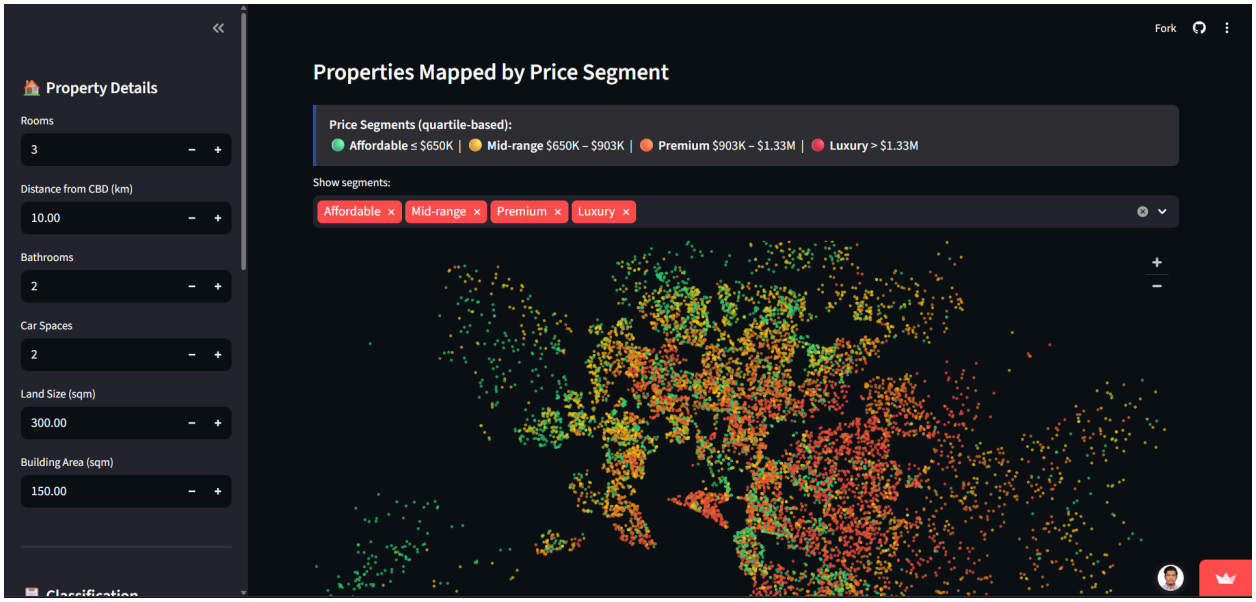
```python
# Path to the saved Gradient Boosting model
model_path = r"D:\SIT720\distinction_task\model\gradient_boosting_model.pkl"

# Load the pipeline
loaded_pipeline = joblib.load(model_path)

# Example input data (replace with your actual feature values)
input_data = {
    'Rooms': 3,
    'Distance': 10,
    'Bathroom': 2,
    'Car': 2,
    'Landsize': 159.0,
    'BuildingArea': 150,
    'Method':'S',
    'SellerG':'Ray',
    'Suburb': 'Abbotsford',
    'Type': 'h',
    'Regionname': 'Northern Metropolitan',
    'CouncilArea':'Yarra'
}

# Convert the dictionary into a DataFrame
input_df = pd.DataFrame([input_data])

# Make predictions
predictions = loaded_pipeline.predict(input_df)

# Display the predictions
print(f"Predicted Housing Price: {predictions[0]:.2f}")
```

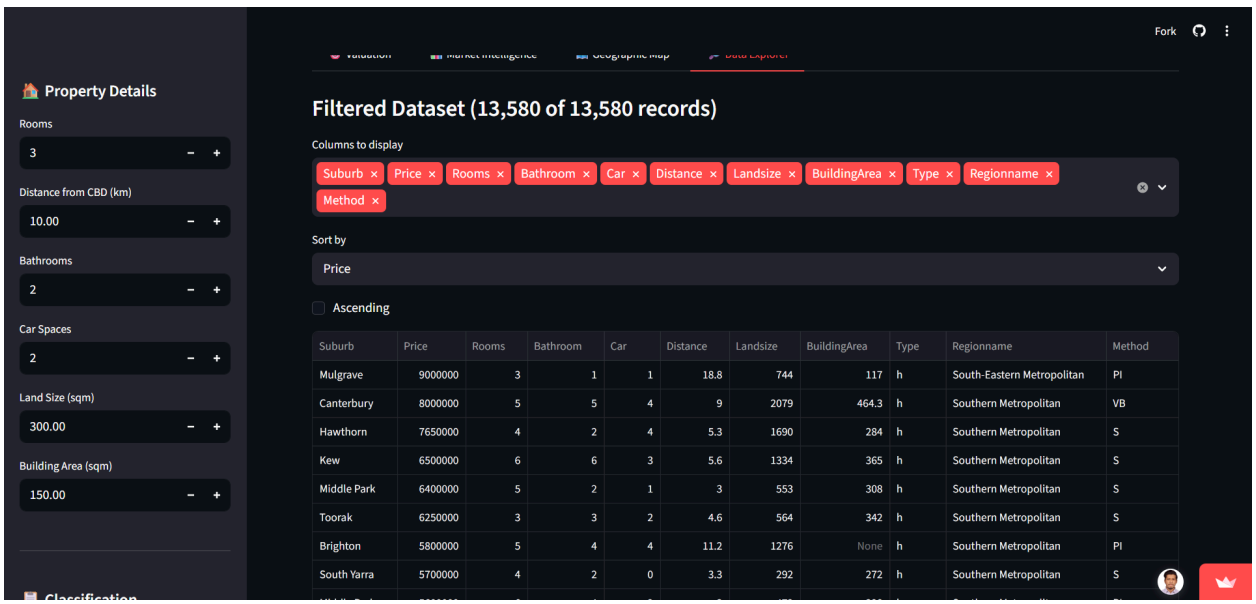
```
Predicted Housing Price: 798506.69
```

This deployment pattern enables stakeholders such as real estate agents, buyers, and investors to query the model without writing code, lowering the barrier to adoption and shortening the feedback loop between modeling and end-user value.


Fork
Property Details
Rooms
3
Distance from CBD (km)
10.00
Bathrooms
2
Car Spaces
2
Land Size (sqm)
300.00
Building Area (sqm)
150.00
Melbourne Housing Intelligence
AI-powered property valuation & market analytics — for buyers, sellers, agents and investors.
LISTINGS
13,580
MEDIAN PRICE
$903K
AVG $/SQM
$7K
REGIONS
8
MODEL R²
0.812
Valuation
Market Intelligence
Geographic Map
Data Explorer
Estimated Property Value
Run Valuation
ESTIMATED MARKET VALUE
$890K
90% Confidence: $801K — $979K
vs Eastern Metropolitan Median
Comparable Sales
Suburb Price Rooms Bathroom Distance BuildingArea
Eaglemont $1,816,000 3 1 10.3 180
Eaglemont $1,601,000 3 1 10.3 136
Eaglemont $1,100,000 3 2 10.3 300
Heidelberg Heights $720,000 3 1 10.5 None
Heidelberg Heights $740,600 3 2 10.5 None
Heidelberg Heights $580,000 3 1 10.5 None
Heidelberg Heights $994,000 3 2 10.5


Fork
Property Details
Rooms
3
Distance from CBD (km)
10.00
Bathrooms
2
Car Spaces
2
Land Size (sqm)
300.00
Building Area (sqm)
150.00
Melbourne Housing Intelligence
AI-powered property valuation & market analytics — for buyers, sellers, agents and investors.
LISTINGS
13,580
MEDIAN PRICE
$903K
AVG $/SQM
$7K
REGIONS
8
MODEL R²
0.812
Valuation
Market Intelligence
Geographic Map
Data Explorer
Market Overview
Median Price by Region
Southern Metropolitan
Eastern Metropolitan
South-Eastern Metropolitan
Northern Metropolitan
Western Metropolitan
Eastern Victoria
$1
$1.01M
$850K
$806K
$793K
$670K
Region
Median Price by Property Type
1.2M
1M
0.8M
0.6M
0.4M
Median
1114 listings
9449 listings
3017 listings





Suburb	Price	Rooms	Bathroom	Car	Distance	Landsize	BuildingArea	Type	Regionname	Method
Mulgrave	9000000	3	1	1	18.8	744	117	h	South-Eastern Metropolitan	PI
Canterbury	8000000	5	5	4	9	2079	464.3	h	Southern Metropolitan	VB
Hawthorn	7650000	4	2	4	5.3	1690	284	h	Southern Metropolitan	S
Kew	6500000	6	6	3	5.6	1334	365	h	Southern Metropolitan	S
Middle Park	6400000	5	2	1	3	553	308	h	Southern Metropolitan	S
Toorak	6250000	3	3	2	4.6	564	342	h	Southern Metropolitan	S
Brighton	5800000	5	4	4	11.2	1276	None	h	Southern Metropolitan	PI
South Yarra	5700000	4	2	0	3.3	292	272	h	Southern Metropolitan	S

12. Conclusion

This project demonstrates a complete, reproducible machine learning workflow—from raw data to interactive deployment— for the task of residential property price prediction in Melbourne.

- **Model performance:** the Gradient Boosting Regressor achieved the strongest R^2 and RMSE among the four candidate algorithms.
- **Robust preprocessing:** median imputation, Z-score outlier removal, standardization, and one-hot encoding produced a clean, model-ready dataset.

- **Practical deployment:** The Streamlit application makes the model directly usable by real estate professionals and prospective buyers.

12.1 Future Work

- **Geospatial enrichment:** incorporate neighborhood-level amenities, school ratings, and transport accessibility.
- **Advanced ensembles:** Evaluate XGBoost, LightGBM, and CatBoost to benchmark against gradient boosting.
- **API deployment:** expose the model via a FastAPI endpoint for integration into third-party platforms.
- **Monitoring:** add drift detection and periodic retraining to maintain accuracy as market conditions evolve.
- **Explainability:** integrate SHAP for per-prediction explanations, improving transparency for end users.

References

11. Kaggle — Melbourne Housing Snapshot dataset: https://www.kaggle.com/datasets/dansbecker/melbourne-housing-snapshot
12. Pedregosa, F. et al. “Scikit-learn: Machine Learning in Python.” JMLR 12, 2011.
13. Friedman, J. H. “Greedy Function Approximation: A Gradient Boosting Machine.” Annals of Statistics, 2001.
14. Streamlit — Official documentation: https://docs.streamlit.io